Assignment_05

Q1. Define hash cryptography. Explain the fundamental principles and techniques of cryptographic hash functions with suitable examples.

Answer:

Definition

Cryptographic hash function is a mathematical algorithm that converts input data of any size into a fixed-size output called a hash value, hash code, or message digest.

It is a one-way function, meaning the original message cannot practically be recovered from its hash.

Fundamental principles

1. Fixed-length output: It produces a hash of a fixed size regardless of the input size.
2. Deterministic: The same input always produces the same hash.
3. Avalanche effect: A small change in input produces a significantly different hash.
4. One-way property: It is computationally infeasible to recover the original input from the hash.
5. Collision resistance: It should be difficult to find two different inputs with the same hash.

Techniques of cryptographic hash functions

- Padding: Extra bits are added to the message to prepare it for processing.
- Block processing: The message is divided into fixed-size blocks.
- Compression function: Each block is processed through mathematical operations to update the internal hash state.
- Finalization: The final internal state produces the message digest.

Example

Suppose we calculate the SHA-256 hash of two messages:

- Input 1: Hello
- Input 2: hello

Both inputs produce 256-bit hashes, but their hash values will be different because the inputs differ.

Conclusion: Cryptographic hash functions are essential for data integrity, authentication and digital signatures.

Q2. Discuss the role of cryptographic hash functions in cryptographic applications. Explain their use in data integrity verification, digital signatures, password storage, and message authentication.

Answer:

Cryptographic hash functions play an important role in protecting information and verifying that data has not been changed.

1. Data integrity verification

A hash is calculated for a file before transfer and compared with the hash calculated after transfer. If the hashes match, this provides evidence that the file has not changed, assuming a secure hash function and trusted reference hash.

Example: SHA-256 is used to verify downloaded software files.

2. Digital signatures

A hash of the message is calculated, and the resulting digest is signed using the sender's private key. The receiver verifies the signature using the sender's public key and checks the message digest.

Example: RSA or ECDSA digital signatures can use SHA-256.

3. Password storage

Websites should not store passwords as plain text. Passwords should be processed using a password-hashing algorithm with a unique salt.

Examples: Argon2id, scrypt, bcrypt and PBKDF2.

A salt helps prevent attackers from using precomputed hash tables. Fast hashes such as plain SHA-256 should not be used alone for password storage.

4. Message authentication

A keyed hash-based mechanism can verify both message integrity and authenticity.

Example: HMAC-SHA-256 uses a secret key and SHA-256 to generate a message authentication code.

Conclusion: Hash functions protect data integrity and support authentication, digital signatures and secure password storage.

Q3. Explain the advantages and disadvantages of hash cryptography. Discuss the major properties of a secure cryptographic hash function, including preimage resistance, second-preimage resistance, and collision resistance.

Answer:

Advantages

1. Provides efficient data integrity verification.
2. Produces a fixed-size digest for data of any size.
3. Supports digital signatures and message authentication.
4. Helps protect stored passwords when suitable password-hashing algorithms are used.
5. Does not require encryption or decryption keys for an ordinary hash function.

Disadvantages

1. A hash alone does not encrypt data or provide confidentiality.
2. Different messages can theoretically produce the same hash.

3. Weak algorithms such as MD5 and SHA-1 are unsuitable for collision-resistant security applications.
4. A plain hash cannot authenticate a message because an attacker may replace both the message and its hash.
5. Fast hash functions are unsuitable for password storage without an appropriate password-hashing scheme.

Major security properties

1. Preimage resistance

Given a hash value h, it should be computationally infeasible to find a message m such that:

$$ H(m)=h $$

Example: Given a SHA-256 digest, an attacker should not be able to feasibly recover its original message.

2. Second-preimage resistance

Given a particular message m_1, it should be infeasible to find a different message m_2 such that:

$$ H(m_1)=H(m_2) $$

Example: An attacker should not be able to create a different file with the same hash as a specified original file.

3. Collision resistance

It should be infeasible to find any two different messages m_1 and m_2 satisfying:

$$ H(m_1)=H(m_2) $$

Example: It should be impractical to create two different documents with the same secure hash value.

Conclusion: A secure hash function should provide strong preimage, second-preimage and collision resistance.

Q4. Describe the Secure Hash Algorithm (SHA) family. Explain the development and key features of SHA-1, SHA-2, and SHA-3, highlighting their major differences.

Answer:

The Secure Hash Algorithm (SHA) is a family of cryptographic hash functions used to generate fixed-length message digests for security applications.

1. SHA-1

- Developed by the NSA and published by NIST in 1995.
- Produces a 160-bit hash value.
- Uses a 512-bit message block.

- Was widely used in digital certificates and digital signatures.
- It is no longer considered secure against collision attacks.

2. SHA-2

- Published by NIST in 2001.
- Includes SHA-224, SHA-256, SHA-384 and SHA-512, among other variants.
- SHA-256 produces a 256-bit hash.
- SHA-512 produces a 512-bit hash.
- Provides strong security when appropriately implemented.

3. SHA-3

- Standardized by NIST in 2015.
- Based on the Keccak algorithm.
- Uses a different internal design from SHA-1 and SHA-2.
- Includes SHA3-224, SHA3-256, SHA3-384 and SHA3-512.
- Provides an alternative design to SHA-2.

Differences between SHA-1, SHA-2 and SHA-3

Feature	**SHA-1**	**SHA-2**	**SHA-3**
Standardized	1995	2001	2015
Main design	Merkle–Damgård	Merkle–Damgård family	Sponge construction
Hash size	160 bits	224–512 bits, depending on variant	224–512 bits, depending on variant
Collision security	Broken	Secure with appropriate variants	Secure with appropriate variants
Current status	Obsolete for security use	Widely used	Secure alternative

Conclusion: SHA-2 and SHA-3 are recommended for modern security applications. SHA-1 should not be used where collision resistance is required.

Q5. Explain the applications of SHA algorithms in cryptographic protocols and information security. Discuss their role in digital certificates, digital signatures, file integrity checking, and secure communication.

Answer:

SHA algorithms are used in modern cryptographic systems to generate message digests and support data integrity and authentication.

1. Digital certificates

Digital certificates contain information about an entity and its public key. Certificate authorities digitally sign certificates, commonly using a signature algorithm combined with a secure hash such as SHA-256.

Example: A certificate may use an RSA-with-SHA-256 or ECDSA-with-SHA-256 signature.

2. Digital signatures

SHA calculates a digest of a message. A digital signature algorithm signs that digest, allowing the receiver to verify the signature and detect message changes.

Example: RSA-PSS with SHA-256.

3. File integrity checking

SHA-256 can generate a digest for a file. Comparing it with a trusted reference digest helps determine whether the file has changed.

Example: Checking the SHA-256 checksum of downloaded software.

4. Secure communication

Secure communication protocols use hash functions and related constructions to protect messages and establish secure connections.

Examples:

- TLS/HTTPS: Uses hash functions in handshake signatures and transcript verification.
- HMAC-SHA-256: Provides message integrity and authentication.
- Key derivation: Hash-based constructions help derive cryptographic keys.

5. Blockchain and other applications

SHA-256 is used in Bitcoin's hashing operations, including block hashing and proof-of-work. Hash functions are also used in software distribution, forensic investigations and data verification.

Conclusion: SHA algorithms are fundamental to modern information security because they support integrity checking, digital signatures, certificate validation and secure communication.